

THE MOLECULAR DIAMETER OF DEUTERIUM AS DETERMINED BY VISCOSITY MEASUREMENTS¹

By A. B. VAN CLEAVE² and O. MAASS³

Abstract

The viscosities of deuterium over the range 23 to -183°C . have been measured. The viscosities of the two-component system deuterium-hydrogen have been measured over the whole concentration range at 22°C . The results show that the deuterium molecule has the same diameter as the hydrogen molecule. The interest attached to the viscosity results for the two-component system are pointed out. The difference between the molecular volume of liquid deuterium oxide and that of liquid water is attributed by the authors to a difference in equilibrium between associated and non-associated molecules in the respective systems.

Experimental

The apparatus used in the viscosity determinations was essentially the oscillating disc type described in detail by Sutherland and Maass (2). A few minor improvements have been made. These will be described in a subsequent paper. This apparatus is ideally suited to the measurement of viscosity coefficients where only small amounts of gas are available. Besides, it has the advantage of affording an exceedingly accurate method of measuring gaseous viscosities at all temperatures below room temperature. It is estimated that the values given below are correct to better than 0.20%, and have a relative accuracy of 0.10%.

Preparation and Purification of Deuterium

The deuterium was prepared from one gram of deuterium oxide (sp. gr. 1.1058), which was purchased from the Ohio Chemical and Manufacturing Company. The deuterium was liberated by allowing the oxide to react with pure metallic sodium. Fig. 1 is a diagram of the apparatus used in the preparation and purification of the deuterium.

The apparatus was first evacuated and filled with dry air. The tip was then cracked off the deuterium oxide container, *D*, which was quickly lowered into the tube *A*. *A* was then immediately sealed off at *C*. By making the tube *A* long, the sealing off could be accomplished without causing condensation of any

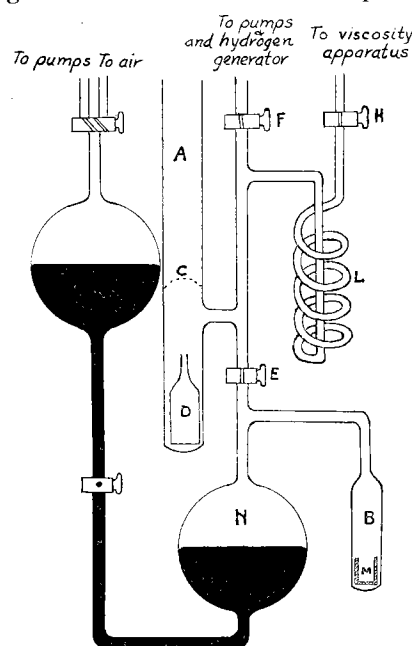


FIG. 1. Diagram of apparatus used in preparation and purification of deuterium.

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Contribution from the Department of Chemistry, McGill University, Montreal, Quebec, Canada.

² Demonstrator, Department of Chemistry, McGill University.

³ Macdonald Professor of Physical Chemistry, McGill University.

water, from the flame, on the inside of the tube. The deuterium oxide was then frozen in liquid air and the whole system evacuated and flushed with pure dry hydrogen. The frozen deuterium oxide was allowed to melt in the presence of the hydrogen to expel dissolved air. Tube *A* was again immersed in liquid air and the system thoroughly evacuated. Tube *B* containing the aluminium cup *M*, nearly filled with metallic sodium, was then surrounded by liquid air while *A* was allowed to warm up to room temperature. When all the deuterium oxide had sublimed into *B* the system was again evacuated and the stopcock *E* closed. The liquid air was then removed from *B*, allowing the reaction to take place.

The deuterium was collected over very pure mercury in the 500 cc. bulb *N*. When *N* was nearly filled with gas the reaction was stopped by replacing the liquid air around *B*. Stopcock *E* was then opened, *H* and *F* being closed. The coil *L* was surrounded by a Dewar flask containing liquid air. Stopcock *H* was then partly opened and the deuterium slowly forced into the evacuated viscosity apparatus. Previous to evacuation the viscosity apparatus had been flushed with pure, dry hydrogen. The reaction in *B* was then allowed to proceed, being aided by heating to 200°C. In this manner enough deuterium was obtained to fill the viscosity apparatus to a pressure slightly greater than atmospheric at room temperature. The gas pressure in the viscosity apparatus could be read on an attached absolute manometer.

The method of making the deuterium-hydrogen mixtures was as follows. Part of the original deuterium was removed from the viscosity apparatus and the pressure noted. Very pure hydrogen was then admitted and the pressure again read, the temperature being maintained constant during this operation. The molecular percentages of the constituents were calculated directly from their partial pressures. All mixtures were allowed to stand for 24 hr. before making any viscosity determinations, thus insuring perfect mixing of the two gases.

Results

Before making any measurements on deuterium, check determinations were made on pure hydrogen prepared by two different methods:— (a) hydrogen from a Kipp generator, purified as indicated by Sutherland and Maass (2); and (b) hydrogen prepared by allowing pure distilled water to react with metallic sodium, as outlined above. In all cases the values for the viscosity coefficients agreed with those given by Sutherland and Maass (2) within 0.16%. This affords an excellent check on the purity of hydrogen prepared by the sodium method.

Since the specific gravity of the original deuterium oxide was 1.1058 it contained 98.0% of D₂O by weight. Hence, on a basis of molecular percentages the deuterium formed 96.08% of the mixture, the remainder being ordinary hydrogen.

Table I is a summary of the results obtained with this mixture. Each value is the mean of three or four independent runs. These results are shown graphically in Fig. 2.

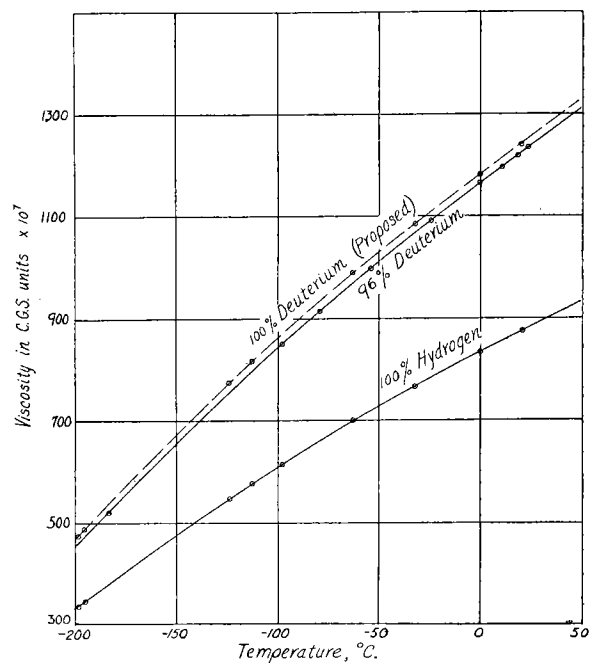


FIG. 2. Variation of viscosity with the temperature.

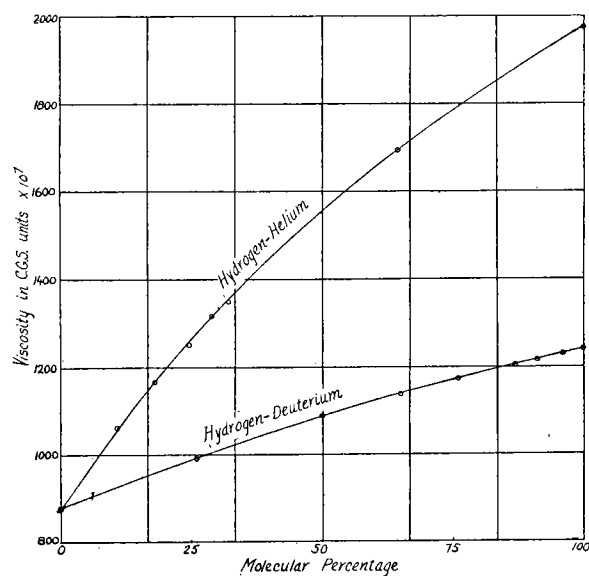


FIG. 3. Variation of viscosity with concentration.

TABLE I
VARIATION OF VISCOSITY WITH TEMPERATURE
(98.0% BY WEIGHT DEUTERIUM, 2.0%
BY WEIGHT HYDROGEN)

Temp., °C.	$\eta \times 10^7$	Temp., °C.	$\eta \times 10^7$
23.61	1234.9	-24.23	1092.1
18.11	1218.9	-53.90	998.7
10.37	1196.3	-79.13	914.7
-0.19	1165.3	-183.00	520.0

results are graphically represented in Fig. 3.

TABLE II
VISCOSITIES OF DEUTERIUM-HYDROGEN MIXTURES

Deuterium, molecular per cent	Temp., °C.	$\eta \times 10^7$	$\eta_{22}^\circ \times 10^7$ (calcd.)	Deuterium, molecular per cent	Temp., °C.	$\eta \times 10^7$	$\eta_{22}^\circ \times 10^7$ (calcd.)
1.0000	22.00	—	1242.5	.6513	22.06	1138.3	1138.2
.9608	22.47	1232.4	1231.0	.4998	21.96	1088.2	1088.3
.9111	22.00	1217.0	1217.0	.2564	22.02	991.3	991.3
.8677	22.07	1205.1	1205.0	.0000	22.00	878.8	878.8
.7634	21.73	1173.6	1174.3				

Discussion

The well known expression

$$\eta = 1/3mnxl$$

can be reduced to $\eta = kM/\sigma^2$ for purposes of this discussion. η is the coefficient of viscosity of a gas, m the mass of one molecule, n the number of molecules in 1 cc. of gas, x the average velocity of the molecules, l the mean free path, M the molecular weight, σ the molecular cross section, and k a constant for a given temperature. Hence, we have

$$\frac{\eta_D}{\eta_H} = \sqrt{\frac{M_D}{M_H}} \cdot \frac{\sigma_H^2}{\sigma_D^2},$$

where the D subscripts refer to deuterium and the H subscripts to hydrogen. If $\sigma_D = \sigma_H$ then

$$\eta_D = \sqrt{\frac{M_D}{M_H}} \cdot \eta_H = \sqrt{2} \cdot \eta_H.$$

The extrapolation of the deuterium-hydrogen curve in Fig. 3 shows that the viscosity of deuterium at 22°C. is $1.414 \pm .002$ times the viscosity of hydrogen at that temperature. Thus, within the writers' experimental error the radii of deuterium and hydrogen molecules at 22°C. are the same.

The ratio $\eta_D : \eta_H$ was found to be constant over a large temperature range. and hence the molecular volumes of deuterium and hydrogen are the same.

Table II shows the result of diluting the deuterium with hydrogen. For comparison all the experimental values have been reduced to their values at 22.0°C. (fourth and eighth columns). As before, each value is the mean of at least three independent runs. The value for 100% deuterium was obtained by extrapolation. These

In this connection it is of interest that the molecular volumes of liquefied D_2O and H_2O are not the same.

Lewis and MacDonald (1) give the specific gravity of pure D_2O at $25^\circ C$. as 1.1056, as against 1.111 calculated by assuming that D_2O has the same molecular volume as ordinary water. This means that the apparent molecular volume of liquid D_2O is greater than that of liquid H_2O .

If this were due to the difference between the volume of the deuterium atom and that of the hydrogen atom then, since the oxygen contributes a large part of the volume in the oxides, the difference would have to be greater than 1%. The viscosity experiments show that as far as the deuterium and hydrogen molecules are concerned a difference of such magnitude does not exist. To account for the difference between the molecular volume of liquid D_2O and that of liquid H_2O , the authors advance the hypothesis that the D_2O and H_2O molecules have the same volume but that in the equilibria



the first is displaced more in the direction of association than the second. In that case the difference in molecular volume ought to become less with rise in temperature, as this favors the formation of the unassociated molecules. Lewis and MacDonald found this to be the case.

Sutherland's equation fails to represent the variation of viscosity with temperature for most gases at low temperatures. The results of Sutherland and Maass (2) for hydrogen and the authors' results for deuterium do not conform to this equation. The relation proposed by Jean's, $\eta/\eta_0 = \left(\frac{T}{T_0}\right)^n$, gives the best representation of the viscosity variation of hydrogen and deuterium over the temperature range of $25^\circ C$. to $-80^\circ C$. With $n=0.699$ the calculated values for the viscosity of 98.0 weight per cent deuterium check exceedingly well with the experimental values over this temperature range. However, at $-183^\circ C$. the deviation of calculated from experimental value is 3.13%. Similar results are obtained when this equation is applied to pure hydrogen.

The whole question of temperature variation of viscosity at low temperatures is an interesting one, and the data accumulated in this laboratory will be published shortly.

From the curves in Fig. 2 it is found that over the temperature range 25° to $-80^\circ C$. the 96 molecular per cent deuterium curve can be obtained from the 100% hydrogen curve by multiplying values on the latter by $1.395 \pm .003$. Below $-80^\circ C$. this relation begins to fail until at $-183^\circ C$. the 96 molecular per cent deuterium has a viscosity of only $1.368 \pm .003$ times that of hydrogen.

The proposed viscosity curve (Fig. 2) for 100% deuterium was constructed by extrapolation. From $25^\circ C$. to $-80^\circ C$. this curve should be very nearly

the correct one. Deviations from the true values will be greater at the lower temperatures. Table III gives values taken from this curve.

The viscosities of gas mixtures are of considerable interest from the point of view of the kinetic theory. Deuterium-hydrogen and deuterium-helium mixtures are of particular importance. Now that it has been shown that the cross section of the deuterium molecule is the same as that of the hydrogen molecule, the approximations of the mean diameters necessary in some theoretical deductions are unnecessary. The first system therefore is one in which molecules of equal diameter but different weight influence the viscosities. In the second system the molecules are of equal weight but have different molecular diameters. The data for the first system are shown graphically in Fig. 3. In this figure the data for the hydrogen-helium system,

TABLE III
VISCOSITIES OF DEUTERIUM AT
DIFFERENT TEMPERATURES

Temp., °C.	$\eta \times 10^7$	Temp., °C.	$\eta \times 10^7$
20	1240	-100	861
0	1181	-120	787
-20	1120	-140	710
-40	1060	-160	632
-60	997	-180	550
-80	931	-200	468

as determined by Trautz and co-workers (3, 4), are given for purposes of comparison. The authors intend to obtain the data for the deuterium-helium system, and when this is done the relations deduced for mixtures by Trautz, Chapman and others will be critically examined. Preliminary calculations on the system deuterium-hydrogen seem to show that

further amplification of the theory is necessary.

References

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